

Pre-empting Multi-Agent Risks from Advanced AI through the AI Action Plan

Executive Summary

The Cooperative AI Foundation welcomes the Administration's focus on making the most of the opportunities available from rapid advances in AI. As the United States works to advance global leadership in AI innovation, a critical emerging consideration involves pre-empting risks that arise specifically from interactions between multiple AI systems.

America's competitive edge in AI will increasingly depend not only on developing powerful individual AI systems but on ensuring these systems work effectively together. As AI systems become more autonomous and are deployed at scale, they will inevitably interact with each other across sectors like finance, transportation, healthcare, and energy. This transition from isolated AI systems to interconnected networks of AI agents offers tremendous opportunities for economic growth and productivity, but also introduces unique challenges that require thoughtful management.

Our research indicates that certain multi-agent risks, if left unaddressed, could significantly impede private sector innovation by creating market inefficiencies, as well as leading to system failures, undermining consumer trust, and generating unnecessary regulatory backlash. Strategic government coordination in targeted areas can help prevent these outcomes without a heavy-handed approach to AI governance.

We recommend strategic investments in research, technical standards for interoperability, and market mechanisms that enable AI systems to coordinate effectively. These targeted measures will help prevent problems like AI collusion in markets, miscoordination in critical infrastructure, and security vulnerabilities in multi-agent systems while creating a more stable environment for private sector innovation.

By proactively addressing these multi-agent risks, the United States can:

- Create a competitive advantage in developing advanced AI agents that function reliably in complex environments
- Prevent market failures that could trigger more restrictive regulations
- Establish global standards that benefit U.S. companies
- Accelerate the safe deployment of productivity-enhancing AI agents across the economy

In this submission, we outline the specific failure modes and risk factors unique to multi-agent AI systems, along with targeted policy recommendations that support the Administration's goals. Our approach emphasizes the importance of addressing these challenges through coordination, standards, and targeted research.

1. Introduction

The rapid advances in artificial intelligence over the past decade have positioned the United States at the forefront of a technology revolution. As we move from an era of relatively simple AI models to increasingly autonomous AI agents capable of taking actions in the world, we stand on the threshold of extraordinary economic opportunities. The success of this transition will depend not just on the capabilities of individual AI systems, but on how effectively they interact with each other.

The future AI landscape will be inherently multi-agent, meaning that many AI systems, potentially developed by different organizations with different objectives, will inevitably interact with one another. We are already seeing the early stages of this transition: algorithmic trading systems interact in financial markets, recommendation systems compete for user attention, and AI assistants communicate with each other and with other digital systems. Soon, more advanced AI agents will manage critical infrastructure, participate in business operations, and represent individuals and organizations in increasingly complex interactions.

This shift presents both unprecedented opportunities and novel challenges. On one hand, networks of effectively coordinating AI agents could unlock extraordinary productivity gains, enable more personalized services, and create entirely new business models. On the other hand, failures in these multi-agent systems could lead to market inefficiencies, service disruptions, and security vulnerabilities at scales and speeds beyond what we have seen in traditional software systems.

Importantly, these challenges are distinct from those posed by single AI systems. Even if individual AI systems are perfectly aligned with their users' interests and comply with all relevant regulations, their interactions can produce unintended and potentially harmful outcomes. For example:

- AI pricing algorithms may implicitly collude to raise consumer prices, even without explicit instructions to do so
- Autonomous systems in transportation or energy grids may fail to coordinate effectively, leading to inefficiencies or service disruptions
- AI agents representing different organizations may engage in escalating conflicts or arms race dynamics that harm all participants

The private sector alone cannot fully address these challenges because they often involve coordination problems, information asymmetries, and externalities that affect entire markets or sectors. Yet inappropriate regulatory approaches risk stifling precisely the innovation needed to solve these problems.

Instead, the United States has the opportunity to take a leadership role in developing lightweight coordination mechanisms, technical standards, and targeted research initiatives that enable AI systems to interact safely and productively. By addressing these multi-agent risks proactively, we can create an environment where private sector innovation flourishes while avoiding the market failures or public harms that might otherwise trigger overreaching regulatory responses.

This submission draws on the recent technical report *Multi-Agent Risks from Advanced AI*, (Hammond et al., 2025) to identify the key failure modes and risk factors in multi-agent AI systems. It offers concrete recommendations for how the Administration can address them through its AI Action Plan, enabling the benefits of AI innovation while preventing unnecessary barriers.

2. The Emerging Landscape of Multi-Agent AI Systems

While much of the public discourse on AI has focused on the capabilities of individual AI systems, the reality of AI deployment is increasingly multi-agent in nature. Today, multiple AI systems are already involved in tasks ranging from trading million-dollar assets to recommending actions to commanders in military contexts. In the near future, applications will expand to include energy management, transportation networks, critical infrastructure, and personal assistants that interact with each other on behalf of their users.

This transition to multi-agent AI is being driven by several factors:

1. **Competitive advantage through autonomy:** Organizations that deploy autonomous, adaptive agents will have significant advantages over those relying on non-adaptive systems or those requiring constant human oversight.
2. **Network effects:** As more AI agents are deployed, the value of having AI systems that can effectively interact with other AI systems increases dramatically.
3. **Division of labor:** Complex tasks often require specialized expertise, making it more efficient to have multiple specialized agents rather than one generalist system.

These trends are already evident in several domains. In financial markets, algorithmic trading systems now execute the majority of trades, interacting with each other at speeds far beyond human reaction times. In online spaces, AI content moderation systems and recommendation algorithms interact in complex ways across platforms. And we are beginning to see early versions of AI assistants that can interact with other digital systems and services on behalf of users.

As foundation models and agent architectures continue to advance, we should expect the proliferation of more capable AI agents interacting in increasingly complex ways. This is not merely a quantitative change but a qualitative one: the shift from isolated AI systems to interconnected networks of AI agents introduces systemic properties and risks that cannot be understood by examining individual systems in isolation.

Understanding and addressing the unique challenges of multi-agent AI will be essential to realizing its full potential for economic growth, improved quality of life, and national security. By taking a proactive approach to multi-agent risks, the United States can establish itself as the leader not just in developing individual AI capabilities, but in creating the infrastructure and standards for AI systems to interact safely and productively.

3. Multi-Agent Failure Modes

Multi-agent systems can fail in various ways, depending on the objectives of the agents and the intended behavior of the system. The report *Multi-Agent Risks from Advanced AI* (Hammond et al., 2025) identifies three distinct failure modes that emerge in multi-agent settings and can lead to significant risks.

3.1 Miscoordination

Miscoordination occurs when AI agents, despite having mutual and clear objectives, cannot align their behaviors to achieve these objectives. This represents the simplest kind of cooperation failure but can still lead to serious problems.

Even in common-interest settings where agents share identical goals, miscoordination abounds due to several factors:

- **Incompatible Strategies:** When agents independently develop different approaches to solve the same problem, they may end up working at cross-purposes. For example, in autonomous driving, models trained on different driving conventions (such as yielding right versus left for emergency vehicles) can fail catastrophically when interacting on the same roads.
- **Credit Assignment Challenges:** In complex environments with multiple agents, it becomes difficult to determine which agent's actions contributed to positive or negative outcomes, making learning and adaptation more challenging.
- **Limited Communication:** Split-second decisions or situations where communication is too costly can lead to coordination failures, especially in "zero-shot" interactions where agents have no prior history of working together.

3.2 Conflict

In most real-world strategic interactions, AI agents will have objectives that are neither identical nor completely opposed, but mixed. Even if these agents are perfectly aligned with their respective users or deployers, conflicts can arise due to the diverging interests of those users.

Key instances of conflict include:

- **Social Dilemmas:** AI systems may enable actors to pursue selfish incentives more effectively, potentially overcoming the technical, legal, or social barriers that normally help prevent destructive competition. For example, AI assistants could enable "hyper-switching" between services or overconsumption of common resources.

- **Military Domains:** AI systems serving as advisors or negotiators in high-stakes military decisions could lead to rapid unintended escalation if not robustly designed, as demonstrated in research where multiple LLMs controlling simulated nation-states rapidly developed arms race dynamics even from neutral starting conditions.
- **Coercion and Extortion:** Advanced AI systems might enable various forms of coercion through surveillance, hacking, or adversarial attacks on other AI systems, potentially creating new forms of strategic threats.

3.3 Collusion

While cooperation failures represent significant risks, there are also settings where cooperation between AI systems is undesirable. AI collusion refers to unwanted cooperation between AI systems at the expense of other parties.

Prominent instances include:

- **Markets:** AI systems could learn to implicitly coordinate pricing strategies without explicit instructions to do so, as has already been demonstrated in both theoretical models and empirical studies of algorithmic pricing in markets like retail gasoline.
- **Steganography:** Recent research has shown that language models can exchange hidden messages that appear innocuous to overseers but contain covert information, with more advanced models showing greater proficiency in such communication.

These failure modes are particularly concerning because they may become more severe as AI capabilities improve, unlike miscoordination problems that might naturally decrease with greater AI sophistication. Additionally, many promising approaches to ensuring the safety of advanced AI are implicitly multi-agent, such as adversarial training or oversight schemes, which could be undermined by collusion between AI systems.

Addressing these multi-agent failure modes requires approaches that go beyond ensuring the safety and alignment of individual AI systems, as even perfectly aligned individual systems can produce harmful outcomes when interacting.

4. Risk Factors

The *Multi-Agent Risks from Advanced AI* report identifies seven critical risk factors that can lead to or exacerbate the failure modes described previously. These factors are largely independent of the agents' precise incentives and can arise across various multi-agent settings.

4.1. Information Asymmetries

Information asymmetry refers to situations where interacting agents possess different levels of information bearing on a joint action. Despite the information processing capabilities of AI systems, they remain vulnerable to failures caused by information asymmetries in several ways:

- **Communication Constraints:** Space or time limitations can prevent complete information exchange, even when agents share common goals.
- **Bargaining Inefficiencies:** When agents with different objectives negotiate, uncertainty about others' values or capabilities can lead to failed agreements or even costly conflicts.
- **Deception:** Different strategic interests can incentivize AI agents to mislead other agents or manipulate markets, as demonstrated in research showing reinforcement learning agents learning to manipulate financial benchmarks.

These information asymmetries can lead to miscoordination even among cooperative agents, or escalate conflicts among competitive ones.

4.2. Network Effects

As AI systems are integrated into existing networks, new risks emerge from the intricate relationships between individual components and the overall system:

- **Error Propagation:** Information can be corrupted as it moves through networks of AI systems, with factual accuracy degrading through repeated transformations. This is a phenomenon demonstrated in experiments where information accuracy fell from 96% to under 60% after multiple AI-driven rewrites.
- **Network Rewiring:** AI systems may increasingly interact with other AIs rather than humans, potentially creating new patterns of connection with unforeseen consequences for resource distribution or system stability.
- **Homogeneity and Correlated Failures:** The current foundation model paradigm means many AI agents may be powered by a small number of similar underlying models, creating critical nodes in the overall network and introducing correlated risks of shared failure modes.

4.3. Selection Pressures

The evolutionary pressures that shape AI systems (whether through gradient descent, developer choices, or user preferences) can lead to concerning outcomes:

- **Undesirable Dispositions from Competition:** Competitive multi-agent settings may select for conflict-prone dispositions like vengefulness, aggression, or deception,

similar to evolutionary pressures in biological systems.

- **Undesirable Dispositions from Human Data:** Models trained on human data can exhibit biases that either reduce or exacerbate risks of conflict, depending on whether they inherit cooperative or competitive tendencies.
- **Undesirable Capabilities:** Co-adaptation between agents can quickly lead to emergent self-supervised auto-curricula, generating increasingly sophisticated strategies through interaction that may include manipulation or deception.

Experiments have shown significant differences in how different LLM populations maintain cooperation across generations when subject to evolutionary selection pressures.

4.4. Destabilizing Dynamics

When multiple adaptive agents interact, their collective behavior can produce unpredictable and potentially harmful dynamics:

- **Feedback Loops:** The 2010 Flash Crash, where algorithmic trading agents entered an unexpected feedback loop leading to a trillion-dollar market loss in minutes, illustrates how multi-agent systems can produce rapid, destabilizing effects.
- **Cyclic Behavior and Chaos:** Mathematical analysis suggests that as the number of learning agents increases, chaotic dynamics can become the norm rather than the exception, making prediction increasingly difficult.
- **Phase Transitions:** Small changes to system parameters can cause abrupt qualitative shifts in overall behavior, potentially leading to sudden and unexpected failures.
- **Distributional Shift:** The actions of adapting agents create a constantly changing environment for other agents, making it challenging to maintain performance over time.

4.5. Commitment and Trust

The ability to form credible commitments can help overcome cooperation failures but may also introduce new risks:

- **Inefficient Outcomes:** Without trust or commitment ability, agents may fail to reach mutually beneficial agreements, especially in high-stakes situations.
- **Threats and Extortion:** The same commitment mechanisms that enable cooperation can be used to make credible threats, as illustrated by historical examples like

automated nuclear response systems.

- **Rigidity and Mistaken Commitments:** Overly rigid commitments may prevent adaptation to changing circumstances, potentially leading to harmful outcomes when new information arises.

4.6. Emergent Agency

Novel forms of agency can emerge at the collective level that are not present in any individual system:

- **Emergent Capabilities:** Narrow systems for separate tasks could combine to enable complex capabilities beyond what any individual system can do, such as automated workflows for designing dangerous compounds.
- **Emergent Goals:** Even if individual AI systems lack problematic objectives, their combinations may act as a goal-directed collective, potentially pursuing unintended outcomes that no individual system was designed to seek.

4.7. Multi-Agent Security

The interconnection of multiple AI systems introduces new security vulnerabilities and attack vectors:

- **Swarm Attacks:** Decentralized agents can coordinate to overcome defenses that would be effective against individual attackers.
- **Heterogeneous Attacks:** Combining different AI systems with complementary capabilities can overcome safety measures, as demonstrated by research showing that while individual models rarely generate harmful content (less than 3% success rate), combining models with different capabilities increased success rates to 43%.
- **Social Engineering at Scale:** Multiple AI agents could coordinate sophisticated manipulation campaigns that would be more convincing than those from a single source.
- **Vulnerable AI Agents:** AI agents acting as delegates for humans or organizations create new attack surfaces that could be exploited to extract private information or manipulate the agent to take undesired actions.

These seven risk factors interact with and amplify each other, creating complex challenges that cannot be addressed by focusing solely on individual AI systems. Tackling these factors requires approaches that account for the dynamic, interconnected nature of multi-agent AI systems.

5. Policy Recommendations

Addressing multi-agent risks from advanced AI requires strategic policy actions that enable innovation while preventing harmful outcomes. Based on the findings in the *Multi-Agent Risks from Advanced AI* report, we recommend the following priority actions for inclusion in the AI Action Plan.

5.1. Support Targeted Research on Multi-Agent Risks

The government should prioritize funding for research specifically addressing multi-agent risks, focusing on:

- **Evaluation methodologies:** Develop robust ways to test how AI systems perform in multi-agent settings, including their cooperative capabilities, vulnerability to collusion, and behavior when interacting with diverse agent populations.
- **Coordination mechanisms:** Support research on protocols, standards, and techniques that enable AI systems to coordinate effectively even when developed by different organizations with different objectives.
- **Security testing:** Fund research on multi-agent adversarial testing, including how multiple AI systems might work together to overcome safeguards even when individual systems cannot.
- **Monitoring techniques:** Develop tools for detecting emergent behaviors, collusion, and other concerning patterns in networks of AI agents.

These research priorities require relatively modest funding compared to general AI capabilities research but would address critical gaps in understanding of multi-agent risks.

5.2. Develop Infrastructure for AI Agent Interaction

Just as internet protocols enabled the growth of the digital economy, infrastructure for AI agent interaction will be essential for realizing the benefits of multi-agent AI while managing its risks:

- **Agent identification standards:** Support the development of unique identifiers for AI agents to enable tracking, attribution, and reputation systems.
- **Secure interaction protocols:** Establish standards for secure, authenticated communication between AI systems to prevent manipulation and unauthorized access.

- **Testing environments:** Create sandboxed environments where AI agents can be evaluated for their behavior in multi-agent settings before deployment.
- **Tamper-evident logs:** Develop standards for maintaining records of AI agent interactions that cannot be altered, enabling accountability while preserving privacy.

This infrastructure should be developed with industry input to ensure it meets practical needs while establishing baseline security and interoperability standards.

5.3. Establish Mechanisms to Prevent Harmful Outcomes

Light-touch market mechanisms can help prevent harmful multi-agent behaviors while preserving innovation:

- **Detection tools for algorithmic collusion:** Support the development of tools that can identify when AI systems are implicitly colluding in markets, enabling enforcement of existing antitrust laws without imposing new regulations.
- **Circuit breakers for AI systems:** In critical domains like financial markets or infrastructure, establish mechanisms to temporarily pause or roll back AI agent actions when concerning patterns emerge, similar to circuit breakers in stock markets.
- **Liability frameworks:** Clarify how responsibility is assigned for harms caused by interactions between multiple AI systems, providing certainty for businesses while ensuring accountability.

5.4. International Coordination on Standards and Governance

The United States could benefit from leading international efforts to address multi-agent AI risks:

- **Technical standards:** The U.S. will need technical standards for AI agent interoperability, security, and coordination. Leading the way in standard setting could position U.S. approaches as the global default.
- **Information sharing:** Establishing mechanisms for sharing information about incidents involving multi-agent AI systems will improve collective understanding of risks.

5.5. Education and Workforce Development

Build capacity to address multi-agent AI risks through:

- **Interdisciplinary training:** Support education programs that combine expertise in AI with fields like economics, security, and complex systems, developing professionals who understand multi-agent dynamics.
- **Simulation expertise:** Develop talent in modeling and simulating complex multi-agent systems to better predict and mitigate risks.

These recommendations aim to establish the conditions for safe and productive multi-agent AI development while minimizing constraints on innovation. By addressing these issues proactively through targeted research, infrastructure, and coordination, the United States can maintain its leadership in AI while preventing market failures or harms that might otherwise trigger more restrictive approaches.

6. Conclusion

The transition from individual AI systems to networks of interacting AI agents represents both a tremendous opportunity and a novel challenge for American innovation. As we have outlined in this submission, multi-agent risks from advanced AI are distinct from the risks posed by individual systems and require specific attention in the AI Action Plan.

The United States stands at a crossroads. By proactively addressing these multi-agent risks through targeted research, infrastructure development, and light-touch coordination mechanisms, America can create the conditions for unprecedented innovation and economic growth. Conversely, ignoring these risks could lead to market failures, security vulnerabilities, and public harms that might trigger overreaching regulatory responses.

The approach we have recommended aligns with the Administration's goals of sustaining American leadership in AI while preventing unnecessary barriers to innovation. Rather than imposing burdensome requirements, our recommendations focus on enabling technologies, standards, and research that will help the private sector deploy AI agents more confidently, safely, and effectively.

We urge the Administration to incorporate multi-agent considerations into the AI Action Plan, recognizing that America's competitive edge in AI will increasingly depend not just on developing powerful individual systems, but on creating an ecosystem where multiple AI agents can interact safely and productively. By rising to this challenge, the United States can establish itself as the global leader in the next frontier of artificial intelligence, one characterized by networks of AI agents working together to solve complex problems and create unprecedented value.

The Cooperative AI Foundation stands ready to support these efforts through our research, expertise, and international network of collaborators. Together, we can ensure that the benefits of advanced AI are realized fully, safely, and in a manner that draws on American leadership in this critical technology.

Reference

Hammond et al. (2025). *Multi-Agent Risks from Advanced AI*. Cooperative AI Foundation, Technical Report #1.

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